



OCEAN PLANKTON POPULATIONS AS EXCITABLE MEDIA

■ J. E. TRUSCOTT and J. BRINDLEY

Department of Applied Mathematics,
Centre for Nonlinear Studies,
Leeds LS2 9JT, U.K.

Plankton populations undergo dramatic surges. Rapid increases and decreases by a factor of 10 or more are observed, often separated by relatively stable interludes. We propose a description of plankton communities as excitable systems. In particular, we present a model for the evolution of phytoplankton and zooplankton populations which resembles models for the behaviour of excitable media. The parameter dependency of the various “excitable” phenomena, trigger mechanism, threshold, and slow recovery, is clear, and permits ready investigation of the influence of properties of the physical environment, including variations in nutrient fluxes, temperature or pollution levels.

1. Introduction. A feature of plankton populations is the occurrence of rapid population explosions and almost equally rapid declines, separated by periods of almost stationary high or low population levels. These phenomena have been broadly divided into two types, “spring blooms” and “red tides”. Both involve surges of phytoplankton population, which grow by a plant-like photosynthetic process. The processes by which these outbursts are controlled are not clearly understood, and no candidate as a model stands out above the rest. Some of the mechanisms proposed include the availability of trace elements such as iron (Provasoli, 1978) and vitamin B₁₂ (Nishijima and Hata, 1989), the vertical stability of the water column (Cloern, 1991) and salinity.

Spring blooms, as their name implies, occur seasonally, almost certainly induced by changes in temperature or nutrient availability, connected with seasonal changes in thermocline depth and strength, and consequent mixing. Red tides, in contrast, are more localized outbreaks. They most commonly occur in coastal waters, in estuaries and at fronts (Pingree *et al.*, 1975). While not strongly correlated with a particular seasonal change, their occurrence does coincide with regions and times of high water temperature. Phytoplankton populations during tides are often so dense as to cause coloration of the water, giving rise to the name.

The phytoplankton involved in red tide events fall into two broad categories. The first is made up of diatoms and blue-green algae, very small plankton species that exhibit a fast rate of reproduction. The second group comprises

dinoflagellates such as *Chattonella*, *Gymnodinium* and *Prorocentrum*. These are relatively large species of phytoplankton, and, while their rate of reproduction is significantly slower, they are able to swim actively with the aid of flagellae to adjust their depth in the water column. Consequently they can take full advantage of the available light and nutrient conditions. These two groups adopt fundamentally different approaches to exploiting their environment.

Some of the phytoplankton genera commonly involved, especially the dinoflagellates, secrete large quantities of neurotoxins during the red tide. These toxins can cause substantial mortality of fish, while poisons absorbed by shellfish can cause paralysis and death in sea birds and humans (Almeida Machado, 1978). The understanding of red tides is, therefore, of direct relevance to coastal communities and the fishing industry in susceptible regions, and is consequently the focus of much research.

2. Basis of Mathematical Model. Although red tides differ widely in their location, severity and taxonomy, certain characteristics are ubiquitous, and any acceptable mathematical model must reproduce their features. Such a basic model can then be refined to describe the details of any particular phenomenon. We take the following to be the irreducible characteristics of red tide events:

- The existence of two stable or quasi-stable population levels of the "outbreak" organism. The first stable population level is the pre-tide population. In the absence of a tide-triggering event, this population level will remain stable and virtually constant throughout the red tide season of several months. The second population level is the outbreak level. This state is not strictly stable, as the actual population can vary markedly during a red tide event, but it does remain consistently much higher than the quiescent level for periods of weeks or months.
- The occurrence of rapid outbreaks followed by slow relaxation. Once triggered, red tides develop over a period of about a week. Termination of a red tide also appears to be a rapid event, with the population of the outbreak organism falling quickly to low, subquiescent levels. These changes imply a fast timescale, in contrast to a slow timescale characteristic of the life-time of the outbreak. This slow timescale governs the evolution of the factor or factors controlling the outbreak, which may include an increase in predation, nutrient limitation, or the effects of toxins.
- A trigger mechanism. It is clear that the onset of the red tide event requires some sort of trigger. It is upon the character of the trigger mechanism that most discussion and controversy is centred. Among the proposed mechanisms are nutrient or trace element enrichment, temperature rises,

pollution, and variation of vertical stability. This wide range of possible mechanisms suggests that the process is subtle, and, though quantitative observations are sparse, it does not appear that a large change in population is necessarily the result of large changes in the environment.

- **Cyclic nature.** The occurrence of more than one bloom in a season suggests that the features influencing a red tide event are cyclic and return to their original values (Satora and Laws, 1989). This must apply both to the triggering mechanism and to the mechanism which restores the population to its original state.

The characteristics we have described also constitute the main properties of excitable media (Murray, 1990; Grindrod, 1991). An excitable medium possesses stable equilibria which exhibit qualitatively different behaviour according to the character of any initial perturbation magnitudes. Thus a small perturbation might decay rapidly and monotonically, whilst a larger perturbation might initially grow and perform a large excursion before eventually returning to the equilibrium. Some *threshold* value of a controlling parameter distinguishes the two cases, and in many cases changes in a “slow variable” are involved in the return to equilibrium. This sequence of events can be pictured in a phase plane diagram, as shown in Fig. 1 for the system $u' = f(u, v)$ and $v' = g(u, v)$, where v is the slow variable (Tyson and Keener, 1988).

3. The Model. The model we present describes the red tide environment as a spatially uniform system with populations evolving with time. This evolution is represented by ordinary differential equations. The general form of the

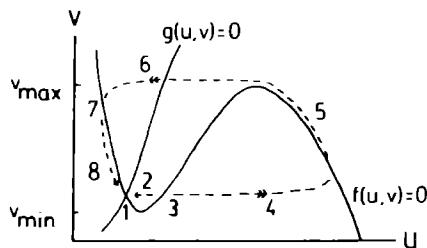


Figure 1. Typical phase plane for an excitable system. The time-dependent variables u and v describe the instantaneous state of the system. Solid lines are null-clines (where $du/dt=0$ and $dv/dt=0$, respectively). Dashed lines are trajectories parameterized by time. The rest state (1) is stable. Small perturbations (2) relax rapidly to rest, but suprathreshold perturbations (3) go on a long excursion before returning to rest. First there is a phase (4) of rapid excitation, during which species u is produced autocatalytically, followed by a period (5) when the system remains in the excited state. After the excitation phase comes to an end (6), the medium is at first absolutely refractory (7) to further excitation and then recovers excitability (8) as it returns to the rest state.

mathematical description can be divided into two parts; the *trigger mechanism*, which controls the initiation of the outbreak, and the *refractory mechanism*, which causes the model to return to its original state.

In any description of red tides as excitable phenomena, the model must contain the idea of two timescales. For the excitability in the phytoplankton population to manifest itself, the timescale for the phytoplankton population change must be much faster than the timescale of the refractory mechanism. If this were not the case, any outbreak would tend to be destroyed by the dynamics of the refractory mechanism and would not be observed. Thus certain resetting mechanisms are not possible within this description. As explained above, there are many possible candidates for the rôle of refractory variable, but these are classifiable into two types; extrinsic and intrinsic.

Extrinsic factors are those not directly used or produced by the phytoplankton themselves, but which nonetheless have an effect on their growth. The phytoplankton do not have any feedback effect on these factors. This group will include such things as salinity, water temperature, and vertical stability of the water column, all of which effect the development of plankton populations, but over which the populations have no influence. The timescale for change in these factors will be tidal or seasonal, much greater than the timescale of the red tide outbreak.

Intrinsic factors are those which have an effect on the phytoplankton population, but whose dynamics are also subject to some degree to the behaviour of that population. These would include such things as nutrients and trace elements, which are consumed by the organisms, or otherwise processed, and also such things as toxins, waste products and predators, whose distribution is affected by the presence of phytoplankton. Within the mathematical model, each interdependence would be represented by a pair of coupled ordinary differential equations; one to represent the phytoplankton and one for the "factor".

The requirement for the phytoplankton and the refractory variable to evolve on a different timescale precludes certain factors from being chosen as the controlling factor. For example, many descriptions (Busenberg *et al.*, 1990) assume a one-to-one ratio between the consumption of nutrient and growth of phytoplankton. Such a description requires both variables to change on the same timescale, and thus nutrient would be unable to act as a refractory variable in this model. There is in any case evidence that red tides normally occur in regions of nutrient excess.

Many possible refractory variables are neither entirely intrinsic nor extrinsic. For example, vitamin B₁₂ is consumed by phytoplankton, but the net source of it is from land run-off and production by bacteria, sources that the phytoplankton do not directly affect (Nishijima and Hata, 1989). A similar situation would presumably hold for iron concentrations.

The essence of the trigger mechanism assumed in our model lies in the interaction of the growth rate of the phytoplankton with the grazing rate of the zooplankton. [There is evidence that herbivore grazing plays a crucial rôle in the initial stages of a red tide outbreak (Wyatt and Horwood, 1973; Levin and Segel, 1976; Uye, 1986).] The mechanism is then modelled in the phytoplankton evolution equation:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right) - R_m Z \frac{P^2}{\alpha^2 + P^2}. \quad (1)$$

Here P represents the population of phytoplankton and Z that of the zooplankton.

Here the gross rate of production of phytoplankton is represented by a logistic growth function, with a maximum growth rate, r , and a carrying capacity, K . Predation of the phytoplankton is represented by a Hollings Type-III function (Holling, 1959), where R_m is the maximum specific predation rate and α governs how quickly that maximum is attained as prey densities increase. The idea of a predation function which saturates for high prey densities is a common one, supported by observation (Uye, 1986).

Figure 2 shows the value of dP/dt against phytoplankton population, P , for typical parameter values [see equations (6)] and Z assumed to maintain the constant value, $5 \mu\text{g N/l}$.

In Fig. 2 can be seen the essence of the trigger mechanism. The lower left-hand zero corresponds to the quiescent population level and the right-hand zero the quasi-steady outbreak population; the central zero is the "trigger" level beyond which the population "explodes" to the upper level. The sensitivity of the model can be gauged from the closeness of the two left-hand zeros. In order for the system to behave in this way, certain features are necessary. Specifically, grazing should exceed production for some interval of phytoplankton population densities, whilst the reverse must be true for very low population densities. If this were not the case, the phytoplankton population would quickly be eradicated. Additionally, saturation of the grazing function is an essential property, as it allows the phytoplankton population to escape from the grazing pressure and form a tide. A crucial point to remember is that the form of dP/dt as shown in Fig. 2 is only strictly valid if all other parameters (except P) are constant in equation (1). If a parameter in equation (1) is dependent on the value of P , then the form of Fig. 2 will change with P . However, since P changes at a much greater rate than any of the dependent parameters in the proposed model, Fig. 2 does give some indication of the initial behaviour of the system. The global behaviour of the system can only be evaluated by considering the null clines of the full system.

All of the properties are found in the Hollings Type-III grazing term. The

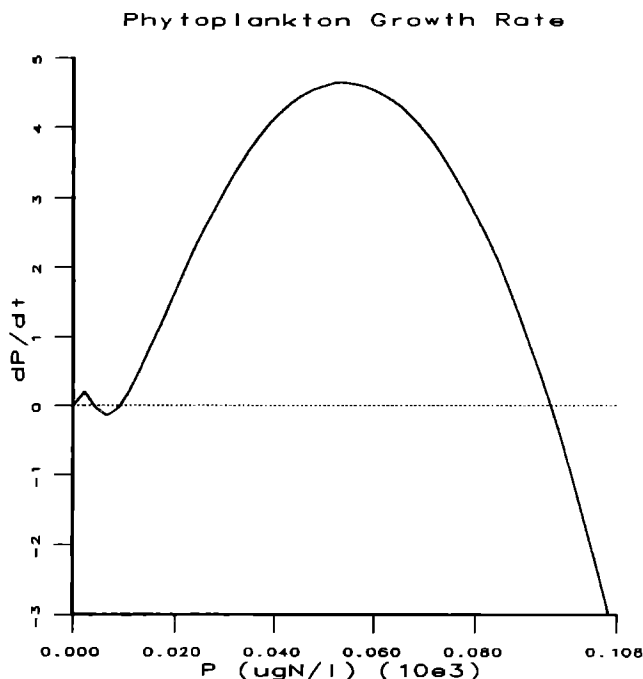


Figure 2. Net rate of production for the parameter values in the text [equations (6)].

form of this grazing function is unusual in that the herbivore's response to low prey density is very small. This suppression of grazing is usually associated with active hunting behaviour on the part of the predator, as opposed to passively waiting to encounter food. It implies that the herbivore has a complicated response to its prey, and to suggest such behaviour for invertebrates may seem a questionable assumption. However, in reality, the raptorial behaviour of copepods is highly complex. Their response to prey shows a marked affinity for particular sizes (Uye, 1986). The interaction of this preference with the size-structured phytoplankton found in the environment could easily lead to seemingly anomalous feeding behaviour. Other evidence (Morey-Baines, 1978) indicates that copepods can exhibit an active hunting behaviour. They are also able to change their feeding patterns by switching prey in conditions of scarcity. These types of behaviour are explicitly identified with the Hollings Type-III grazing function (Ludwig *et al.*, 1978).

The Hollings Type-III term has rarely been used in ecological models, although it is by no means unprecedented. A more common and intuitively obvious choice would be a Hollings Type-II form, such as the Michaelis-Menten grazing function:

$$R_m Z \frac{P}{a + P};$$

or the Ivlev grazing function:

$$R_m Z (1 - e^{-\lambda P}).$$

However, while both these functions could give rise to two stable states, the lower one will be at $P=0$ which implies the extinction of the population. These functions could be made to conform to the qualitative requirements by including a lower cut-off phytoplankton density for the grazing function. For example, in the case of the Ivlev grazing function, $I(P)$:

$$I(P) = \begin{cases} R_m Z (1 - e^{-\lambda(P-P_0)}), & P_0 \leq P \\ 0, & 0 \leq P < P_0. \end{cases}$$

The inclusion of this lower cut-off value is effectively a crude parameterization of the complex feeding behaviour discussed above, but it introduces a new parameter, P_0 , to be estimated or ascertained. In addition, it removes the smoothness of the Hollings Type-III function, which one intuitively feels should be a feature of the "averaged" behaviour of a large group of individuals, as well as rendering any analysis much less tractable.

The refractory factor in the system presented in this paper is the population of zooplankton. The rate of production of zooplankton is controlled by the population density of phytoplankton, while their loss from the system is through death and natural predation by higher members of the food web. The full system is:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right) - R_m Z \frac{P^2}{\alpha^2 + P^2} \quad (2)$$

$$\frac{dZ}{dt} = \gamma R_m Z \frac{P^2}{\alpha^2 + P^2} - \mu Z; \quad (3)$$

where Z represents the population density of zooplankton. The rate of removal of zooplankton by death and predation is modelled as being proportional to the zooplankton population, with specific rate, μ .

The constant γ represents the ratio of biomass consumed to biomass of new herbivores produced. This is admittedly a crude device, since the relationship between these two quantities is unlikely to be linear and probably involves a time-lag between consumption and reproduction. γ covers a wide range of processes in nutrition and reproduction: only a proportion of the zooplankton are capable of reproduction, only some of the food ingested is assimilated and

only a small amount of this is used for reproduction. Fortunately, it is not necessary to calculate these ratios directly as γ can be estimated implicitly from the values of steady population levels (Uye, 1986) and the value of μ (Wake *et al.*, 1991). The presence of γ or a similar effect is essential to the functioning of this model. It is a manifestation of the need of the two equations to evolve on different timescales as described above.

In order to expose clearly the mathematical character of these equations it is convenient to non-dimensionalize them, using the substitutions:

$$P = K\tilde{P}, \quad Z = K\tilde{Z} \quad \text{and} \quad t = \frac{\tilde{t}}{R_m}.$$

This non-dimensionalization sets the maximum sustainable outbreak at unity, ensuring that all dependent variables remain $O(1)$ throughout the process, and exposing in an explicit way the relationship between the timescales for phytoplankton population change and the refractory mechanism (in this case the zooplankton population).

Introducing:

$$v = \frac{\alpha}{K}, \quad \beta = \frac{r}{R_m}, \quad \text{and} \quad \omega = \frac{\mu}{\gamma R_m};$$

we obtain, dropping the tildes, the system:

$$\frac{dP}{dt} = \beta P(1 - P) - Z \frac{P^2}{v^2 + P^2} = f(P, Z), \quad (4)$$

$$\frac{dZ}{dt} = \gamma \left(\frac{P^2}{v^2 + P^2} - \omega \right) Z = g(P, Z). \quad (5)$$

Typical results from the literature (Uye, 1986; Wake *et al.*, 1991) correspond to parameter values:

$$\begin{aligned} K &= 108 \mu\text{g N/l}, \quad r = 0.3/\text{day}, \quad R_m = 0.7/\text{day} \\ \alpha &= 5.7 \mu\text{g N/l}, \quad \mu = 0.012/\text{day}, \quad \text{and} \quad \gamma = 0.05; \end{aligned} \quad (6)$$

which yield:

$$\beta = 0.43, \quad v = 0.053, \quad \text{and} \quad \omega = 0.34. \quad (7)$$

Equations (4) and (5) have equilibria when (P, Z) satisfy:

$$f(P, Z) = g(P, Z) = 0;$$

that is, at $(0, 0)$, $(1, 0)$ and $(P_0, Z_0) \simeq (0.035, 0.046)$. Of these, $(0, 0)$ represents

the total absence of both species, $(1, 0)$ the equilibrium population of P in the absence of Z , and (P_0, Z_0) is the stable pre-outbreak co-existing state. The stability of these solutions is discussed in an Appendix to this paper.

Although these parameters have been tuned to give particular values for the equilibrium state of P and Z , the underlying mathematical structure of the system is very robust.

Figure 3 shows the form of the null clines, $f(P, Z)=0$ and $g(P, Z)=0$. For the system to exhibit excitable behaviour, it is necessary that the null cline, $f(P, Z)=0$, retains its characteristic shape, i.e. that $f(P, Z)=0$ should have two turning points at values of P greater than zero (in Fig. 3, the local minimum occurs at $P=P_H$). The positions of these turning points are the solutions of:

$$\frac{\partial f}{\partial P}=0$$

$$f(P, Z)=0. \quad (8)$$

Substituting $f(P, Z)$ directly into equation (8) and eliminating Z gives the turning points as solutions of

$$P^3 - \frac{P^2}{2} + \frac{v^2}{2} = 0. \quad (9)$$

Clearly the position of the turning points, such as P_H in Fig. 3, are functions of v only. Equation (9) can have either one or three real roots. To retain the excitability of the system, there must be at least two turning points and therefore the cubic must have three real roots. By considering the number of

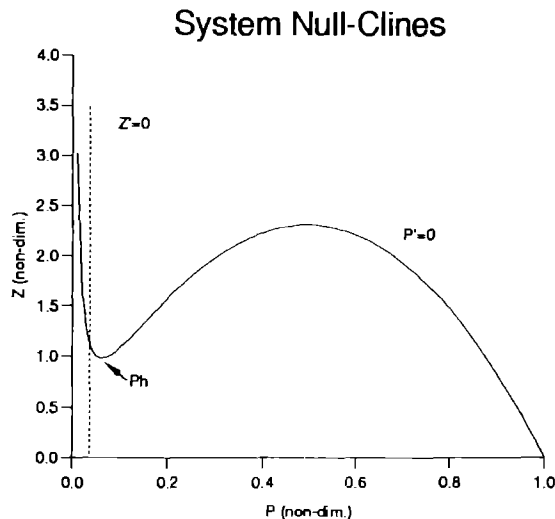


Figure 3. Null-clines for the parameter values in the text [equation (6)].

real roots of equation (9), it can be shown that this occurs only within the parameter range,

$$0 < v < \frac{1}{3\sqrt{3}}.$$

This condition is satisfied by a range of realistic parameter values.

The null cline, $g(P, Z)=0$, occurs on the line $P=P_z$, where:

$$P_z = v \sqrt{\frac{\omega}{1-\omega}}. \quad (10)$$

The system is capable of exhibiting two different types of behaviour, dependent upon the position of P_z with respect to the two roots of equation (8). When P_z falls below the lower or above the upper root, the system behaves excitably. With P_z falling between the two roots, the system has no stable equilibrium and solutions follow a periodic trajectory. It can easily be shown that Hopf bifurcations separate these regions of parameter space (see Appendix). The red tide outbreaks are modelled with the system in region of parameter space defined by:

$$P_0 < P_H. \quad (11)$$

As the non-dimensionalization makes clear, the qualitative behaviour is governed by v and ω only and is quite independent of the parameter, γ , or the ratio of maximal birth-rate to predation rate for the phytoplankton, β .

Outbreaks can be stimulated in various ways, for example:

- (i) by direct perturbation. This may not seem a credible method of stimulation for an ecological system. However, there is strong evidence linking the occurrence of red tide outbreaks with the presence of "seedbeds" (Iwasaki, 1989). These seedbeds contain populations of dormant dinoflagellates which under the appropriate conditions can become active again, rapidly increasing the effective population.
- (ii) A more realistic trigger mechanism is a change (increase) in r , the maximum growth rate. In reality, this growth rate is very sensitive to temperature changes; for example, the growth rate of *Chattonella Antiqua* roughly doubles between 20 and 25°C (Uye, 1986). Looking only at the response of the phytoplankton, the system behaves as follows. As r increases, the two lower zeros of Fig. 2 move together until they merge and disappear; P is now attracted to the single remaining equilibrium, corresponding to an outbreak. The effect of the refractory mechanism is to "pull down" the graph shown in Fig. 2. As Z increases, the ratio of grazing rate to growth rate increases. First the two lower

solutions of $P'=0$ are restored. Continued increase in herbivore population causes the upper two solutions of $P'=0$ to draw together and disappear. At this point, the phytoplankton population collapses back to the region of lower equilibrium. The subsequent decay of the grazing population restores the graph of P' back to its appearance in Fig. 2. In terms of the full system as described by the null-clines in Fig. 3, the perturbation corresponds to an increase in β . This effectively "raises" the $P'=0$ null-cline and shifts the Z coordinate of the stable state. The size and rate of change of β dictates whether the system returns to equilibrium directly or by an extended trajectory as shown in Fig. 1: in other words, whether an outbreak occurs or not.

- (iii) Other possible methods of triggering include a reduction in either the zooplankton population, Z , or in their predation efficiency. Either of these could lead to the coalescence and disappearance of the two zeros in Fig. 2. Such a reduction could be linked to a sudden increase in external predation on Z , or possibly to the effects of pollution, which could be responsible for a reduction in both the Z population and their efficiency as predators.

We give numerical examples of some of these cases in Section 4.

4. Numerical Results. Numerical integration of equations (2) and (3) is straightforward, and we present here some particular examples using, except where noted, the parameter values of Satora and Laws (1989):

- (1) The simplest numerical expedient is to stimulate an outbreak by direct perturbation of P . As outlined above, this could be seen as representing the effect of the activation of dormant phytoplankton cells already present in benthic seedbeds. Figure 4 shows the result of such a perturbation of the phytoplankton population beyond the triggering point. The two populations eventually return to their equilibrium values.
- (2) Increase in birth-rate of phytoplankton. An increase in the birth-rate, r , of phytoplankton will cause the two lower equilibria in Fig. 2 to merge and disappear. Clearly, dP/dt is now always positive and P must increase to the remaining equilibrium, corresponding to an outbreak. This is shown in Fig. 5. In this case, the system at equilibrium for $r=0.4$, was perturbed by changing r to 0.6 over 4 days.

Since the qualitative form of the trajectory depends on the magnitude of the perturbation, different rates of change for r would not necessarily be expected to trigger similar response. Numerical simulation seems to confirm this. To model an increase in rate of production for phytoplankton, the parameter, r , was taken as a function of time:

$$r = \min\left(r_0 + \frac{dr}{dt} t, r_{\max}\right);$$

where $r_0 = 0.4/\text{day}$ and $r_{\max} = 0.6/\text{day}$.

As dr/dt is decreased, the response of the system undergoes a marked qualitative change, from excitable behaviour to a slow change of equilibrium position. The onset of this transition occurs over quite a narrow range of dr/dt , centred at $0.005/\text{day}^2$. The location of this transition region was not particularly sensitive to changes in γ , the response rate of the predator community, as might have been expected. This would seem to suggest that for an outbreak to be triggered in this manner there is a minimum "activating" rate of change for the phytoplankton reproduction rate. Consider, as described above, a situation in which an outbreak is triggered solely by a rise in temperature causing an increase in the rate of reproduction of phytoplankton. From Uye (1986), we can calculate a rough value for the rate of change of productivity with temperature, dr/dT :

$$\frac{dr}{dT} \simeq 0.06/(\text{day } ^\circ\text{C}).$$

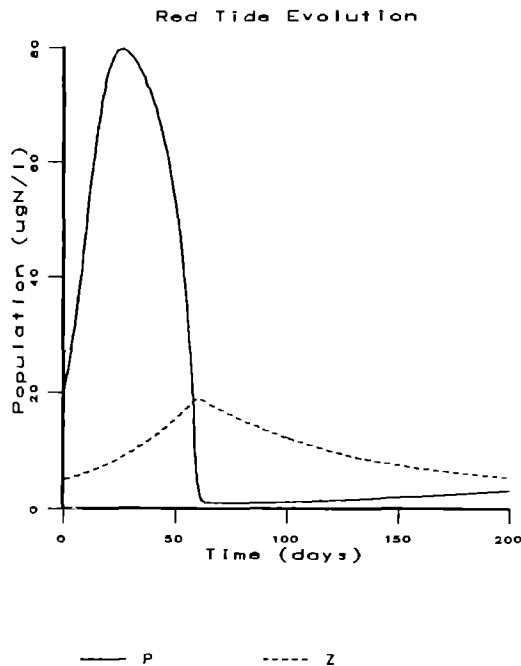


Figure 4. Evolution from non-equilibrium initial state, $(P, Z) = (20, 5)$.

Combining this with critical rate of change value gives an estimate for the minimum rate of change of water temperature which could trigger an outbreak as about $0.08^{\circ}\text{C}/\text{day}$. This mechanism could be particularly relevant to the occurrence of spring blooms.

- (3) Other possible triggering mechanisms. The effect of any of these physical processes is found to be qualitatively similar to that observed in case (2) above, as the two lower equilibria in Fig. 2 coalesce and disappear. The details of course differ according to differences in parameter values, and the consequences of the existence of competing effects, for example increasing temperature and increasing predation efficiency, will need careful numerical study. Additionally, the nature of the carrying capacity, K , is not known precisely, but must have some dependence on nutrient and light levels and toxin production by the organisms themselves.

The results of the simulation, with red tides lasting for about a month, fall within the bounds of observed data (Uye, 1986; Iizuka *et al.*, 1989; Park *et al.*, 1989) and lend credence to this model as a basis for some more detailed examination of the effects of a variety of physical influences.

One quantity often considered in the context of predator-prey interactions is the specific predation rate. This is the fractional change in the phytoplankton

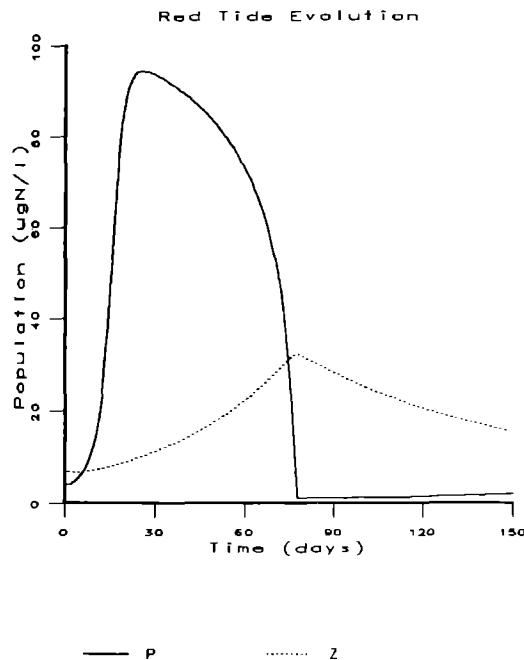


Figure 5. Evolution of phytoplankton population after perturbation of r .

numbers per unit time, and effectively illustrates the impact of predation on the population at any particular time. It is interesting to examine the specific predation rate for the system as the outbreak advances. Figure 6 shows the evolution of a typical outbreak population of phytoplankton, along with the specific predation rate experienced by that population.

In the initial stages of the outbreak, the specific predation rate drops dramatically, illustrating the phytoplankton community's effective escape from predation. The very large population combined with the slow response of the predators means that predation has a very small influence at the peak of the attack, as suggested in the literature (Uye, 1986). It is only as the tide begins to decay that the effects of predation reassert themselves. The large "spike" at the tail end of the tide is caused by two competing effects. As the predator population rises, their effect on the dwindling phytoplankton population increase rapidly. This continues until the phytoplankton numbers drop to such an extent that the predators can no longer graze them efficiently and the specific predation rate drops accordingly.

The appearance of the specific predation "spike" is a product of a system whose refractory variable is the population of predators, and its presence or absence in a real ecosystem would be a measure of the appropriateness of this particular controlling mechanism. However, the pronounced decrease in specific predation as the tide develops is a feature of models of this general type and compares favourably with observational data.

All these possible triggering mechanisms leave unchanged the excitable nature of the system. However, additional predation on Z (i.e. larger μ), or a reduction in grazing efficiency (i.e. smaller γR_m), each lead to an increase in ω .

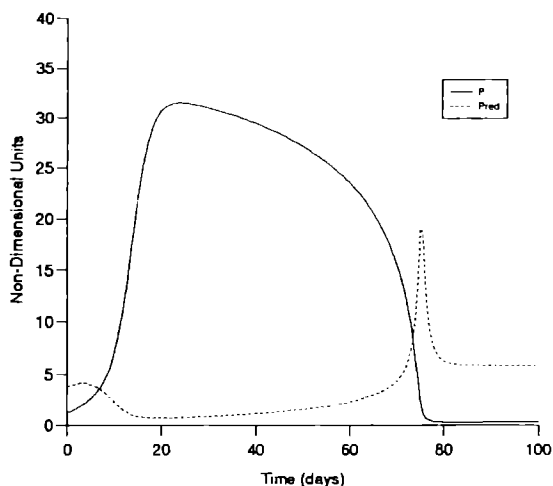


Figure 6. Evolution of specific predation rate during red tide event.

As ω increases, condition (10) is broken. The nature of the unexcited state changes from steady to oscillatory, and the single red tide event is replaced by a periodic sequence of outbreaks as shown in Fig. 7.

5. Discussion. Our model has yielded results which, in a *qualitative* sense, match all the requirements postulated in Section 2. Moreover, when we “calibrate” the model by use of observed steady-state plankton populations, and use the consequent inferred parameter values, we obtain *quantitative* results which are entirely reasonable. The model’s “excitable” behaviour is robust, and the rôle of each of the four parameters, β , ε , ν and ω , is clear. In terms of physical observables, changes in value of parameters may arise through a variety of different processes. For example, ω may be lowered by a reduction in the death rate of zooplankton (μ) or by an increase in the effectiveness of their grazing (γR_m), or again ε may be changed either by changes in γR_m or in the phytoplankton reproduction rate, r . Thus characteristics of the behaviour, for example the triggering of explosive growth, lifetime of tide or bloom, or period for recovery, depend in different ways on the various physical influences, such as temperature change, nutrient or pollutant influence. Such influences may be readily treated within the model and both qualitative and

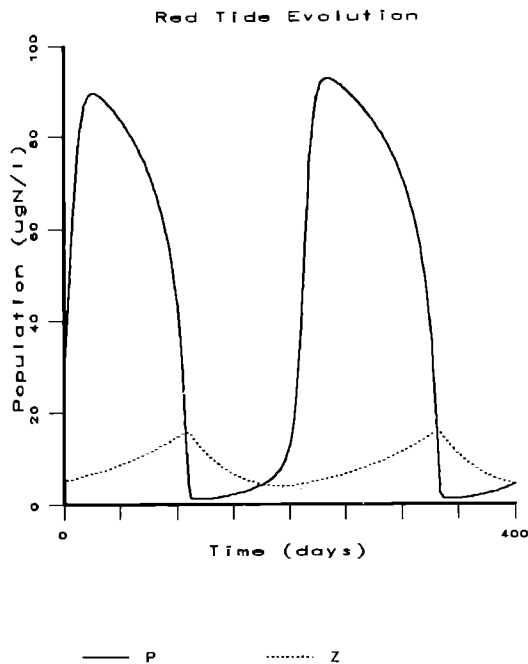


Figure 7. Oscillatory behaviour resulting from breaking inequality (10). Parameter values as in text with the exception that $\mu = 0.024/\text{day}$.

quantitative changes analysed and comprehended; we have included several examples in Section 4.

We have considered only an essentially one-dimensional "well-mixed" situation. Comparison with real plankton distributions and more useful predictions require the introduction of spatial structures, both horizontal and vertical, in which the strongly stratified character of the ocean is likely to have significant effect. Extensive investigations of two- and three-dimensional phenomena have been carried out in excitable media models, usually motivated by biological problems. Corresponding phenomenology in plankton populations might be expected, and ongoing work is exploring these ideas.

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APPENDIX

A.1. Stability Analysis of the System The stability of the various equilibria identified in Section 3 can be ascertained by examining the eigenvalues of the stability matrix, $\mathbf{J}$.

$$\mathbf{J} = \begin{pmatrix} \frac{\partial f}{\partial P} & \frac{\partial f}{\partial Z} \\ \frac{\partial g}{\partial P} & \frac{\partial g}{\partial Z} \end{pmatrix}$$

where the elements of the matrix are the partial derivatives of f and g in equations (4) and (5).

$$\frac{\partial f}{\partial P} = \beta(1 - 2\beta) - \frac{2ZPv^2}{(nu^2 + P^2)^2},$$

$$\frac{\partial f}{\partial Z} = -\frac{P^2}{v^2 + P^2},$$

$$\frac{\partial g}{\partial P} = \frac{2\gamma ZPv^2}{(nu^2 + P^2)^2},$$

$$\frac{\partial g}{\partial Z} = \gamma \left(\frac{P^2}{v^2 + P^2} - \omega \right).$$

The resulting values of $\mathbf{J}$ at the three equilibrium points are shown below. For $(P, Z) = (0, 0)$,

$$\mathbf{J} = \begin{pmatrix} \beta & 0 \\ 0 & -\omega\gamma \end{pmatrix}.$$

In this case the eigenvalues of $\mathbf{J}$ are clearly β and $-\omega\gamma$, indicating that $(0, 0)$ is a saddle point and unstable.

For $(P, Z) = (1, 0)$,

$$\mathbf{J} = \begin{pmatrix} -\beta & -\frac{1}{v^2+1} \\ 0 & \gamma\left(\frac{1}{v^2+1} - \omega\right) \end{pmatrix}.$$

For this matrix, the eigenvalues are $-\beta$ and $\gamma((1/v^2 + 1) - \omega)$. It is easy to show using equation (10) that the second eigenvalue is greater than zero, provided $P_0 < 1$. The point (1, 0) is therefore another saddle point, and unstable.

The eigenvalues of $\mathbf{J}$ at (P_0, Z_0) cannot be obtained by simple inspection. They can be expressed as the solutions of the characteristic equation,

$$\lambda^2 - \text{Trace } \mathbf{J}\lambda + \text{Det } \mathbf{J} = 0. \quad (\text{A1})$$

From the expressions for the derivatives above, it can be seen that $\text{Trace } \mathbf{J} = \partial f / \partial P$, and $\text{Det } \mathbf{J} > 0$ at (P_0, Z_0) . The real parts of the eigenvalues of $\mathbf{J}$ are therefore both of the same sign. The equilibrium is consequently not a saddle point, but either stable or unstable depending on the sign of $\partial f / \partial P$. Values of (P_0, Z_0) at which $\partial f / \partial P = 0$ are points at which the eigenvalues of $\mathbf{J}$ have zero real part, i.e. Hopf bifurcations. These points also fulfil the conditions in equation (8), and are therefore also the turning points of the null-cline $f(P, Z) = 0$.

For the values of the parameters given in the text, $\partial f / \partial P < 0$, giving eigenvalues with negative real parts and a stable equilibrium.

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